SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - Viva Based Questions

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 –** Viva Based Questions	**Assessment:**	**Assessment Tool 3-** Viva Based Questions
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	MURALI KRISHNAN N	**Reg. No.:**	192524090
Section / Batch:	B.TECH AI&DS/ 25 BATCH	**Date:**	20/08/26

Code of Conduct

I MURALI KRISHNAN N (192524090)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____________________________

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

		reasoning and insights				
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

A. Introduction to Statistical Inference

1. **What is statistical inference?**

Answer: Statistical inference is the process of using sample data to draw conclusions about a larger population.

It helps us estimate unknown values and make decisions based on available evidence.

2. **Why is statistical inference important in data science?**

Answer: Statistical inference helps data scientists make reliable conclusions from limited data.

It supports prediction, decision-making, experimentation, and understanding uncertainty.

3. **What is the difference between descriptive statistics and inferential statistics?**

Answer: Descriptive statistics summarize and present the data that has been collected.

Inferential statistics use sample data to make conclusions or predictions about a population.

4. **What is a population in statistics?**

Answer: A population is the complete set of individuals, objects, or observations being studied.

For example, all customers of a company can form a population.

5. **What is a sample?**

Answer: A sample is a smaller group selected from a population for analysis.

It is used to learn about the population without studying every member.

6. Why do data scientists usually work with samples instead of complete populations?

Answer: Studying an entire population can be expensive, time-consuming, or practically impossible.

A properly selected sample can provide useful information with fewer resources.

7. What is a population parameter?

Answer: A population parameter is a numerical value that describes a characteristic of the entire population.

Examples include population mean, proportion, and variance.

8. What is a sample statistic?

Answer: A sample statistic is a numerical value calculated using observations from a sample.

Examples include sample mean, sample proportion, and sample variance.

9. Give a real-world example where statistical inference is used in industry.

Answer: A company may survey a sample of customers to estimate overall customer satisfaction.

The result can help the company make decisions about products or services.

10. What are the two major activities involved in statistical inference?

Answer: The two major activities are point estimation and hypothesis testing.

Point estimation estimates parameters, while hypothesis testing evaluates claims about populations.

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

Answer: The frequentist approach uses repeated sampling and probability to make conclusions from data.

It focuses on the long-run performance of statistical procedures.

12. **What is the basic idea behind frequentist statistics?**

Answer: Frequentist statistics treats the population parameter as fixed but unknown.

The observed sample data are considered to arise from a random sampling process.

13. **How does the frequentist approach differ from the Bayesian approach?**

Answer: In the frequentist approach, parameters are fixed and data are treated as random.

In the Bayesian approach, uncertainty about parameters is represented using probability distributions.

14. **In the frequentist approach, what is considered fixed: the parameter or the data?**

Answer: The population parameter is considered fixed but unknown.

The sample data are considered random because they can vary from one sample to another.

15. **What does long-run frequency mean in frequentist statistics?**

Answer: Long-run frequency is the proportion of times an event occurs over many repeated trials.

It is used to interpret probabilities in the frequentist approach.

16. **Why is random sampling important in the frequentist approach?**

Answer: Random sampling gives each suitable population member a known chance of selection.

It helps reduce selection bias and supports valid statistical inference.

17. **Can a population parameter be considered a random variable in the frequentist approach? Why?**

Answer: No, a population parameter is not considered a random variable in frequentist statistics.

It is treated as a fixed value whose actual value is unknown.

18. **Give an example of a frequentist problem in a data science application.**

Answer: A data scientist may test whether a new recommendation algorithm increases user engagement.

Sample data are used to determine whether the observed improvement is statistically significant.

C. Point Estimation

19. What is a point estimate?

Answer: A point estimate is a single numerical value used to estimate an unknown population parameter.

For example, the sample mean can be used as a point estimate of the population mean.

20. What is an estimator?

Answer: An estimator is a formula or rule used to calculate an estimate from sample data.

For example, the sample mean is an estimator of the population mean.

21. What is the difference between an estimator and an estimate?

Answer: An estimator is the general formula or procedure used for estimation.

An estimate is the actual numerical value obtained after applying the estimator to a sample.

22. What is the point estimator for the population mean?

Answer: The sample mean, written as $\bar{x}$, is the point estimator for the population mean μ.

It is calculated by adding all sample observations and dividing by the sample size.

23. How do you estimate a population proportion from a sample?

Answer: The population proportion is estimated using the sample proportion, written as $\hat{p}$.

It is calculated as the number of successes divided by the total sample size.

24. What is the point estimate of population variance?

Answer: The sample variance, written as s^2, is commonly used to estimate population variance σ^2.

It measures the spread of observations around the sample mean.

25. Why is the sample mean commonly used to estimate the population mean?

Answer: The sample mean uses all observations in the sample and is an unbiased estimator of the population mean.

It also becomes more reliable as the sample size increases.

26. **What properties should a good estimator have?**

Answer: A good estimator should ideally be unbiased, consistent, and efficient.

These properties help the estimator produce accurate and stable estimates.

27. **What is an unbiased estimator?**

Answer: An estimator is unbiased when its expected value is equal to the true population parameter.

Thus, it does not systematically overestimate or underestimate the parameter.

28. **What is meant by the bias of an estimator?**

Answer: Bias is the difference between the expected value of an estimator and the true population parameter.

An estimator with zero bias is called an unbiased estimator.

29. **What is the difference between bias and variance of an estimator?**

Answer: Bias measures the systematic difference between an estimator and the true parameter.

Variance measures how much the estimator changes across different random samples.

30. **Why can two different samples produce different point estimates for the same population parameter?**

Answer: Different random samples usually contain different observations from the same population.

Therefore, their calculated statistics, such as sample means, may also be different.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

Answer: A point estimate is calculated from a sample rather than the entire population.

A different random sample may produce a different estimate of the same parameter.

32. **What is the sampling distribution of an estimator?**

Answer: The sampling distribution is the probability distribution of an estimator over many possible samples.

It shows how the estimator varies from sample to sample.

33. **What is the standard error?**

Answer: Standard error measures the typical amount of variability in a sample statistic across repeated samples.

A smaller standard error generally indicates a more precise estimate.

34. **How is standard error different from standard deviation?**

Answer: Standard deviation measures the spread of individual observations in a dataset.

Standard error measures the spread or variability of a sample statistic.

35. **How does increasing the sample size affect the standard error of the sample mean?**

Answer: Increasing the sample size generally decreases the standard error of the sample mean.

For independent observations, the standard error is approximately $\sigma/\sqrt{n}$.

36. **Why does variability in an estimate decrease when the sample size increases?**

Answer: A larger sample contains more information about the population.

This reduces the effect of random sampling variation on the estimate.

37. **What is the relationship between standard error and confidence intervals?**

Answer: The standard error is used to determine the margin of error in a confidence interval.

A larger standard error generally results in a wider confidence interval.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

Answer: No, different sample means can occur naturally because of random sampling variation.

Both samples can be correct even though they give different estimates.

E. Confidence Intervals

39. What is a confidence interval?

Answer: A confidence interval is a range of values used to estimate an unknown population parameter.

It provides an estimate together with information about sampling uncertainty.

40. Why is a confidence interval more informative than a point estimate?

Answer: A point estimate gives only one numerical value for the parameter.

A confidence interval gives a range and shows the uncertainty associated with the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

Answer: If the same sampling procedure were repeated many times, about 95% of the resulting intervals would contain the true parameter.

It does not mean there is a 95% probability that one particular fixed interval contains the parameter.

42. What is the difference between a confidence level and a confidence interval?

Answer: The confidence level is the long-run coverage percentage of the statistical procedure, such as 95%.

The confidence interval is the actual numerical range calculated from a particular sample.

43. What factors determine the width of a confidence interval?

Answer: The width depends mainly on sample size, data variability, and the selected confidence level.

Higher variability and higher confidence generally make the interval wider.

44. How does increasing the sample size affect the width of a confidence interval?

Answer: Increasing the sample size generally makes the confidence interval narrower.

This happens because the standard error decreases as the sample size increases.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

Answer: Increasing the confidence level from 95% to 99% makes the interval wider.

A wider range is needed to provide greater long-run coverage.

46. What is the difference between a z-confidence interval and a t-confidence interval?

Answer: A z-confidence interval uses the standard normal distribution, while a tconfidence interval uses the t-distribution.

The t-interval is commonly used when the population standard deviation is unknown.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

Answer: The t-distribution is generally used when the population standard deviation is unknown.

It is especially useful for small samples when estimating a population mean.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Answer: Hypothesis testing is a statistical method used to evaluate a claim about a population.

It uses sample evidence to decide whether there is sufficient evidence against a null hypothesis.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

Answer: The null hypothesis H_0 represents the default claim, often indicating no effect or no difference.

The alternative hypothesis H_1 represents the competing claim that an effect or difference exists.

50. How can a confidence interval be used to make a decision about a hypothesis?

Answer: A confidence interval can be compared with the value stated in the null hypothesis.

If the hypothesized value falls outside the interval, H_0 can be rejected at the corresponding significance level.